



AeroPulse-NG

TACTICAL RADAR & AIRSPACE SURVEILLANCE ENGINE

Air-gapped · offline-first · NCAA / ICAO Annex 5 SI-metric

National Drone & Aerospace Innovation Expo 2026

National Innovation Pitch Challenge — Flight Planning & Air Traffic
Management

THE PROBLEM

Nigeria's low-level airspace is a surveillance gap

■ Radar is uneconomical at scale

Primary/secondary installations cost millions of dollars each and demand specialist maintenance.

■ Coverage is uneven

Low-level corridors, secondary aerodromes and terrain-shadowed sectors operate with minimal or no radar picture.

■ Weather degrades exactly when it matters

Harmattan haze and convective storms hit visibility during diversion surges.

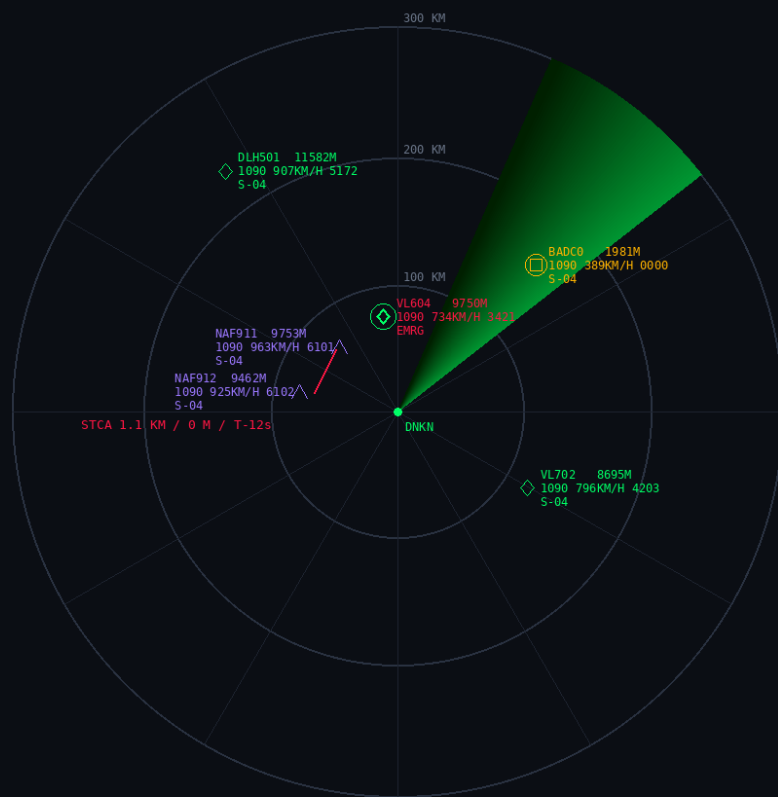
■ Non-cooperative traffic is invisible

Transponder-off aircraft — smuggling, insecurity, rogue drones — defeat dependent surveillance.

When connectivity or a radar head fails, controllers lose the traffic picture precisely when risk peaks.

THE SOLUTION

A radar workstation from USD 150 of radio



TACTICAL PICTURE — LIVE

● SDR-1 1090MHz	412 msg/s
● SDR-2 ACARS	2.1 msg/s
● SDR-3 GUARD	offline
● AWOS RS-485	QNH 1013.2
● EKF ENGINE	248 us

ZULU 12:41:07 WAT 13:41:07

THREAT MATRIX

VL604 50K 7700 - EMERGENCY
NAF911 <-> VL604 - STCA PAIR
BADCO - DARK TARGET

- Software-defined radios in, tactical picture
- 1090 MHz Mode S ADS-B decode with CL
- 6-state Kalman filtering and dead-reckoning
- Automatic conflict alerts (STCA) to ICAO
- Triple-fusion weather: AWOS + ACARS +
- 100% offline — sovereign by construction

HOW IT WORKS

One decode path — synthetic or real RF

AeroPulse-NG — System Pipeline

air-gapped · offline-first · SI-metric presentation (NCAA / ICAO Annex 5)

RF SOURCES

SDR-1 1090 MHz ADS-B / Mode S
SDR-2 131.55 MHz ACARS
AWOS mast RS-485 (sidecar)

SIMULATOR (bench mode)

6-aircraft truth fleet
encodes genuine DF17 frames
same decoders as real RF

HARDWARE DECODE LAYER

mode_s_decoder DF17 · CRC-24 repair
CPR even/odd -> lat/lon
Gillham alt · velocity
acars_decoder framing · CRC-16
METAR / D-ATIS classify
sdr_registry serial-lock channels

KINEMATICS

6-state EKF [x y z vx vy vz]
Joseph-form covariance update
Mahalanobis gate $\chi^2(3)=16.27$
dead-reckoning coasting
leader-line projection 120 s

DISPLAY 1 — RADAR CANVAS

60 fps PPI · rings 100/200/300 km
MIL-STD-2525D symbols
3-line FDB tags (SI units)
flashing STCA connectors 2 Hz
R&B ruler · geofence overlay

SURVEILLANCE ENGINE (60 Hz)

queue > EKF predict > fuse pos/vel
STCA scan 15 Hz (R*-tree)
Doc-4444: 9.26 km / 305 m / 120 s
anomaly scan 1 Hz · geofence 2 Hz
EngineSnapshot {tracks-FDB-STCA-wx}
tokio bus -> telemetry://snapshot

WEATHER FUSION

AWOS > D-ATIS > BDS 4.4/4.5
IDW spatial blend (vector wind)
ISA fallback + confidence
Harmattan dust-layer estimate

DEFENCE OVERLAYS

7700/7600/7500 alerting
dark-target & silence detect
polygon geofences + bands
lead-pursuit intercept solver

DISPLAY 2 — OPERATIONS HUD

flight-strip bay (arr / dep)
triple-fusion weather matrix
FFT diagnostics · latency gauge
threat matrix · intercept calc

PYTHON SIDECAR (NDJSON stdio)

DuckDB flight recorder (batched)
track positions · alerts · weather
AWOS serial reader (+ simulated mast)

Safety-critical math, proven by tests

DECODE

DF17 CRC-24 single-bit repair
Global CPR even/odd solve
Gillham altitude codec
ACARS CRC-16 + METAR/D-ATIS

KINEMATICS

6-state EKF, Joseph-form
update
Mahalanobis gate
 $\chi^2(3)=16.27$
R*-tree spatial index (full)
STCA 9.26 km / 305 m / 120 s

WEATHER

BDS 4,4/4,5 register codecs
ISA atmosphere + IDW blend
Vector-averaged wind
Harmattan dust-layer model

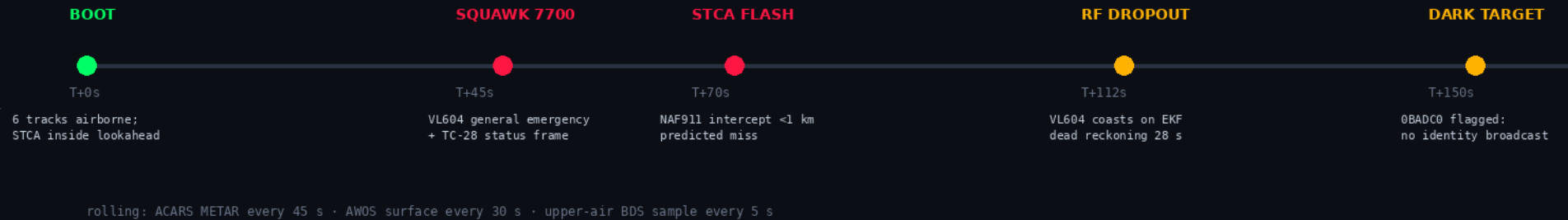
DEFENCE

7700/7600/7500 alerting
Dark-target & silence detection
Polygon geofences + bands
Lead-pursuit intercept solver

104 automated checks green: Rust 90 · Python 14 · TypeScript strict · production bundle

A scripted airspace you can watch

Demonstration scenario timeline (synthetic feed)



■ Judges watch a conflict develop in real time

Emergency squawk, intercept geometry, RF blackout and dark-target flags — every alert path exercised in 3 minutes.

Three radio sources, zero internet

Triple-Fusion Weather Matrix



■ Harmattan-aware

Visibility-driven dust-layer estimate tuned to West African dry-season operations.

■ Honest degradation

Every sample carries source provenance and a confidence score.

WEATHER MATRIX

IDW spatial blend · vector-average wind (never degrees)
ISA fallback with confidence metric (source diversity x range)
priority: live AWOS overrides D-ATIS seeds; METAR fills gaps
dust-layer top estimated from visibility + thermal bump

NCAA / ICAO compliant by construction

Units	ICAO Annex 5 / NCAA — SI metric presentation	km · m · km/h · hPa
Surveillance	DO-260B Mode S / ADS-B MOPS	DF17, CPR, Gillham
Separation	ICAO Doc 4444 radar minima	9.26 km / 305 m / 120 s
Symbology	MIL-STD-2525D + HF-STD-010A	diamond/caret/quad frames
Identity	ICAO Annex 10 squawk semantics	7500 / 7600 / 7700
Integrity	NDAIE rulebook — AI use declared	original work, MIT/Apache deps

What changes if this ships

>99%

capital cost reduction per
surveillance station
(USD 150 receiver vs
multi-million radar head)

24/7

conflict alerting at
aerodromes that can
never justify conventional
radar

100%

sovereign: no cloud, no
internet, no
foreign dependency in the
loop

Open

engineering reference
platform for Nigerian
avionics talent — civil +
military dual-use

Operational: NAMA civil ATC + NAF tactical command share one picture.

Economic: sector-wide low-level coverage becomes a line item, not a capital programme.

Working prototype today, RF tap next

- **Now — verified prototype**

Full engine + dual-display application; 104 automated checks; live demo scenario.

- **Next — real RF front-end**

RTL-SDR I/Q tap feeding the existing decoder (interfaces already shaped).

- **Then — field trial**

Physical AWOS mast, Comm-B weather interrogation, station hardening.

- **Always — open engineering reference**

Documentation suite shipped with the codebase.

Integrity declaration: developed with declared generative-AI assistance per NDAIE rules; all third-party components MIT/Apache licensed. Team roster submitted via official form.

garba-the-analyst · github.com/garba-the-analyst/aeropulse-ng